

Titanic Exploratory Data Analysis (EDA)

CodeAlpha Data Analytics Internship

Task 2: Exploratory Data Analysis

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Introduction:

This report summarizes an Exploratory Data Analysis (EDA) of the Titanic Passenger Dataset. The project explores the dataset structure, cleans missing values, analyzes passenger characteristics, identifies trends and patterns, tests hypotheses, and documents data quality issues.

Objectives:

- Explore the dataset structure.
- Understand variables and data types.
- Handle missing values.
- Identify trends, patterns and anomalies.
- Test hypotheses using statistics and visualization.
- Detect data issues and provide conclusions.
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Dataset:

Dataset: Titanic Passenger Dataset (891 records, common training dataset).

Main variables include Passenger ID, Survived, P class, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin and Embarked.

Data Cleaning:

Missing values in Age and Embarked were handled, Cabin was removed because it contained many missing values, and data types and duplicates were checked.

Key Findings:

- Female passengers showed a higher survival rate.
- First Class passengers had higher survival than lower classes.
- Higher fares were generally associated with better survival.
- Missing values and outliers were identified before analysis.

Conclusion:

The project demonstrates the complete EDA workflow required for the internship, including data exploration, cleaning, visualization, hypothesis testing and interpretation of results.

References

- Kaggle Titanic Dataset